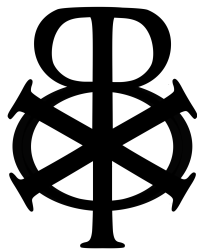


SYNCUSSION SY-1M

BUILD DOCUMENT and BILL of MATERIALS

rev. 1.20

PSYCO X



KIT CONTENTS

- 1 x Voice Board Immersion Gold (ENIG) PCB
- 1 x Control Board Immersion Gold (ENIG) PCB
- 1 x Powder Coated / Screen Printed Metal Enclosure
- 2 x 10k Slide Potentiometers (Audio Taper)
- 6 x 100k Slide Potentiometers (Audio Taper)
- 6 x 100k Slide Potentiometers (Linear Taper)
- 2 x 500k Slide Potentiometers (Audio Taper)
- 16 x Slide Potentiometer Caps
- 1 x Pre-programmed ATMEGA328 MCU
- 1 x 47uH RF Choke
- 2 x 22 pin, press fit, breakaway header strips
- 7 x M3 x 3mm screws
- 4 x Adhesive Rubber Feet

Bill Of Materials

Resistors

1Ω 2W	x 1
100R	x 8
220R	x 1
560R	x 2
680R	x 4
1k	x 57
2k2	x 3
4k7	x 14
5k6	x 1
10k	x 29
22k	x 10
27k	x 2
33k	x 11
47k	x 46
56k	x 4
100k	x 99
150k	x 3
220k	x 16
330k	x 6
470k	x 6
1M	x 6

Mouser Part (Click to View)

[660-SPRX2C1R0J](#)
[603-CFR-25JT-52-100R](#)
[603-CFR-25JB-52-220R](#)
[279-CFR25J560R](#)
[603-CFR-25JR-52-680R](#)
[603-CFR-25JR-521K](#)
[603-CFR-25JR-522K2](#)
[603-CFR-25JR-524K7](#)
[603-CFR-25JR-525K6](#)
[603-CFR-25JR-5210K](#)
[603-CFR-25JR-5222K](#)
[603-CFR-25JR-52-27K](#)
[603-CFR-25JR-52-33K](#)
[603-CFR-25JR-5247K](#)
[603-CFR-25JR-5256K](#)
[603-CFR-25JR-52100K](#)
[603-CFR-25JR-52150K](#)
[603-CFR-25JR-52220K](#)
[291-330K-RC](#)
[603-CFR-25JR-52470K](#)
[603-CFR-25JR-521M](#)

Trimmers

100k x 7

[652-3362P-1-104LF](#)

Capacitors (Film & MLCC)

15p	x 1
22p	x 2
100p	x 1
1n	x 15
2n2	x 2
10n	x 3
22n	x 2
33n	x 2
47n	x 2
100n	x 4
100n	x 3
470n	x 1

[80-C316C150J1G](#)
[810-FG18C0G1H220JNT6](#)
[810-FG18C0G1H101JNT6](#)
[871-B32529C102J189](#)
[871-B32529C222J189](#)
[871-B32529C103K289](#)
[871-B32529C223J289](#)
[871-B32529C1333J289](#)
[871-B32529C1473J189](#)
[871-B32529C1104J189](#)
[810-FG18X7R1H104KNT6](#)
[810-FG24X7R1E474KNT6](#)

Capacitors (Al Electrolytic)

u47/16V	x 4	647-USP1HR47MDD
1u/50V	x 10	647-USR1H010MDD
10u/16V	x 9	UMA1C100MDD1TP
22u/16V	x 1	ECE-A1VKA220
100u/16V	x 1	ECE-A1CKA101
100u/16V	x 2	MAL202190538E3
220u/16V	x 5	75-516D227M016MM6AE3

Capacitors (Tantalum)

u47/35V	x 4	80-T491B474K035
2u2/35V	x 2	74-TR3B225K035C2500

Semiconductors & IC's

1N4148 (Diode)	x 17	512-1N4148
BC547 (NPN)	x 29	512-BC547BBU
BC557 (PNP)	x 24	512-BC557BTA
BCM847 (Matched NPN)	x 18	771-BCM847DS135
BCM857 (Matched PNP)	x 8	771-BCM857DS135
BF256B (JFET)	x 2	512-BF256B
CD4069 (Hex Inverter)	x 6	595-CD4069UBE
LM1458 (Dual Op Amp)	x 4	926-LM1458N/NOPB
RC4458 (Op Amp)	x 9	595-RC4558P
TL082 (Dual Op Amp)	x 3	595-TL082CP
MCP4822 (12 bit DAC)	x 2	579-MCP4822-E/P
6N137 (Opto Isolator)	x 1	859-6N137
TL497 (Switching Reg)	x 1	595-TL497AIN or 595-TL497ACN
LM7808 (8V Reg.)	x 1	511-L7808CV
78L05 (5V Reg.)	x 1	511-L78L05CZ
3mm LED (Red)	x 5	604-WP710A10ID

Switches

12 Way Rotary Switch	x 2	105-CK1049 (you will need to file a flat)
OFF-ON	x 4	642-MHS122K
ON-OFF-ON	x 4	642-MHS233K or 642-MHS123K
(The 4 way DIP Switch is included in the kit)		

Rotary Potentiometer

10k (9mm Rotary)	x 2	858-P090L01T15BR10K
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Connectors

14 pin (2 x7) female header	x 4	517-929975-01-07-RK
3 pin female header	x 2	517-929974-01-03-RK

Connectors cont.

4 pin (2x2) female header	x 2	517-9602046202AR
3.5mm Stereo jack socket	x 5	502-35RAPC2BH3
1/4" Mono jack socket	x 2	550-10201
DC Barrel jack socket	x 1	502-RAPC712X

Miscellaneous components

16MHz Crystal	x 1	449-LFXTAL058383BULK
Ferrite Bead	x 2	81-BL01RN1A1F1J
220uH Inductor	x 1	815-AIUR-05-221K
IC sockets 28 pin	x 1	575-1104732841001000
IC sockets 14 pin	x 1	575-1104731441001000
IC sockets 8 pin	x 17	575-1104730841001000

(There are an additional 6x 14 pin IC Sockets included in the kit)

Hardware

5mm M3 M/F Spacer	x 7	855-R30-3000502
20mm M3 F/F Spacer	x 7	855-R30-1002002
12mm M3 M/F Spacer	x 7	855-R30-3001202
12mm M3 F/F Spacer	x 2	855-R30-1011202
M3 x 5mm Screws	x 15	534-9191-3
Knob for Osc. Mode Switch	x 1	Any 20mm knob of your choice (6mm shaft)
12V DC 300mA Wall wart	x 1	2.5mm centre pin

Tools and Supplies

Most builders will already have the required tools and supplies in their arsenal, but for those new to Synth DIY, this list describes everything you might need in order to complete a successful build.

- Soldering Iron (Preferably temp. controlled. Check the required temperature for your chosen solder)
- Solder (I use Loctite Crystal400)
- Liquid Flux Dispensing Pen (such as Chemtronics CW8100)
- Soldering Tip Cleaner (Keep it clean and tinned !)
- Fine Tweezers (Invaluable for the SMD transistors and capacitors)
- De-soldering Pump (Hopefully, you won't need it)
- Digital Multimeter (DMM) (Always check resistors before fitting!)
- Masking tape (For holding things in place and marking)
- Small Flat Screwdriver (For trimming)
- Posi Screwdriver (For the final assembly)
- Small Flat File (To create a flat side on the Rotary switch shaft)
- Focus and a little patience (Have these, and you're good to go!)

Build Guide

One of the most time consuming aspects of building even a sparsely populated PCB, can be that of merely seeking out individual component locations. In order to speed up the process there are, towards the end of this document, colourised location maps of many (although not all) components. Each component section of the guide has direct links to the relevant maps. Click on the page number in the “Goto” link to jump to the relevant map.

If, in the build list the component name is not colour highlighted, there is no map and the locations should be easy to find.

Components are listed in suggested order of fitting.

Be sure to follow standard precautions when handling static sensitive parts.

Voice Board

SMD Transistors.

8 x **BCM857** Goto [p17](#)

18 x **BCM847** Goto [p17](#)

I would suggest fitting the all of BCM847's first followed by the BCM857's (or the other way round if you prefer). Either way, don't get them mixed up because they aren't easy to distinguish.

Both components have 180° rotational symmetry and can therefore be fitted in either orientation.

If you are new to SMD soldering, check out some of [John Gammell's YouTube videos](#) for expert technique. Better still, get somebody to solder the SMD parts for you!

Check for solder bridges with your Loupe (or other magnifier) before moving on.

SMD Caps (Tantalum)

***Be sure to align the dark band on the device with the marking on the silk screen ***

2 x **u47/35V** Goto [p18](#)

Resistors

2 x **560R** Goto [p18](#)

48 x **1K** Goto [p18](#)

8 x **4K7** Goto [p19](#)

Resistors cont.

18 x	10K	Goto	p19
8 x	22K	Goto	p19
4 x	33K	Goto	p20
40 x	47K	Goto	p20
70 x	100K	Goto	p20
2 x	150K	Goto	p21
6 x	220K	Goto	p21
4 x	330K	Goto	p21
6 x	470K	Goto	p22
4 x	1M	Goto	p22

Diodes

14 x	1N4148	Goto	p22
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(The dark line on the diode must be aligned with the line on the silk screen!)

IC Sockets TIP! - Use masking tape to hold sockets in place while your turn the board over.

6 x	14 pin	Included in the kit
6x	8 pin	

Trimmers

2 x	100k	(VC02 Offset)
-----	------	---------------

Film Capacitors

14 x	1n	Goto	p23
2 x	2n2	Goto	p23

TO-92 Transistors

10 x	BC547	Goto	p24
20 x	BC557	Goto	p24

Aluminium Electrolytic Capacitors

8 x	1u/50V	Goto p25
4 x	10u/16V	Goto p25
4 x	220u/16V	Goto p25

Connectors

4 x	3.5mm Mono jack sockets	(CV and Trigger inputs)
2 x	14 pin female pin sockets	(Board inter-connect)
1 x	4 pin female pin socket	(Board inter-connect)
1 x	3 pin female pin socket	(Board inter-connect)
2 x	1/4" Mono jack sockets	(Audio Outputs)

Potentiometers

2 x	10k 9mm Rotary	(Trigger Sensitivity)
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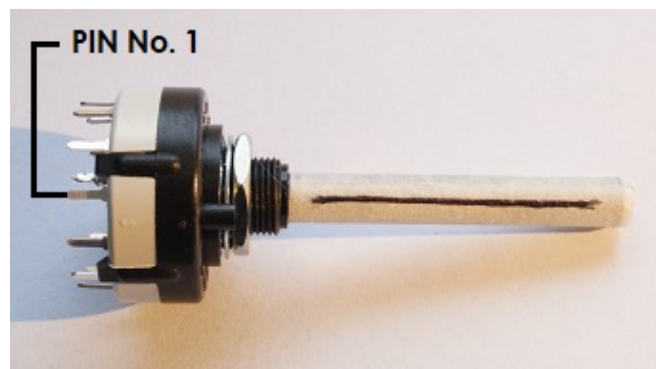
Switches

2 x	12 Way Rotary	(Oscillator Mode Selector)
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BEFORE FITTING -

First of all the stop washer needs to be set to the correct position. To do this, remove the hex nut, locking washer and stop washer. Turn the switch **fully anti-clockwise** until the last click and then re-fit the stop washer with the small lug inserted into the slot labelled "6". Refit the locking washer followed by the hex nut. Check that there are 6 positions (5 clicks in either direction)

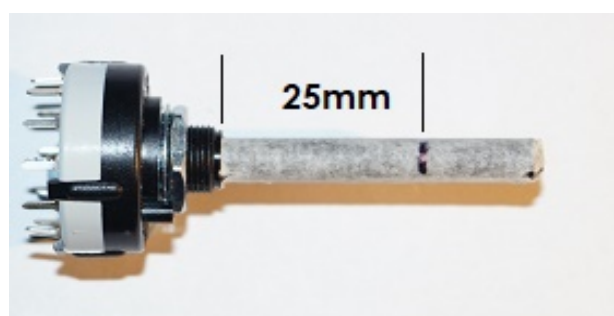
To prevent the switch knob from turning on the shaft you will need to file a flat. With the switch rotated **fully anti-clockwise** wrap a piece of masking tape around the shaft and mark it in line with Pin No. 1



File the line away to create the flat. You don't need to file much, just enough for the cap lug screw to sit flat against and stop the cap from turning on the shaft.



The shaft will also require cutting down. To achieve the correct length, wrap a piece of masking tape around the shaft and mark it at 25mm from the top of the bushing.



Grip the end of the shaft in a bench vice and cut with a small hacksaw.

The switch is now ready for fitting.

IC's (MAKE SURE TO GROUND YOURSELF WHEN HANDLING STATIC SENSITIVE DEVICES)

**There is a slight error on the Silk Screen.
The two location s for the 4558 Op Amps read "4458"**

6 x	CD4069	Hex Inverters	Goto p26
2 x	4558	Dual Op Amp	Goto p26
4 x	1458	Dual Op Amp	Goto p26

That completes the build for the Syncussion SY-1M Voice Board.

Control Board

SMD Capacitors

2 x	u47/35V	Goto p27
2 x	2u2/35V	Goto p27

Resistors

6 x	100R	Goto p28
1 x	220R	Goto p28
4 x	680R	Goto p28
9 x	1K	Goto p29
3 x	2K2	Goto p29
6 x	4K7	Goto p29
1 x	5K6	Goto p30
11 x	10K	Goto p30
2 x	22K	Goto p30
2 x	27K	Goto p31
7 x	33K	Goto p31
6 x	47K	Goto p31
4 x	56K	Goto p32
29 x	100K	Goto p32
1 x	150K	Goto p32
10 x	220K	Goto p33
2 x	1M	Goto p33
2 x	Ferrite Beads	Goto p33

Diodes

3 x	1N4148
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IC Sockets (Make sure the notch on the socket is aligned with the silk screen)

1 x 28 pin DIL Socket

1 x 14 pin DIL Socket

13 x 8 pin DIL Socket

Trimmers

5 x 100k (2 x CV – V/OCT, 2x MIDI – V/OCT, 1x NOISE Trim)

Film Capacitors

1 x 1n Goto [p34](#)

3 x 10n Goto [p34](#)

2 x 22n Goto [p34](#)

2 x 33n Goto [p35](#)

2 x 47n Goto [p35](#)

4 x 100n Goto [p35](#)

MLCC Capacitors

1 x 15p Goto [p36](#)

2 x 22p Goto [p36](#)

1 x 100p Goto [p36](#)

3 x 100n Goto [p37](#)

1 x 470n Goto [p37](#)

Transistors

19 x BC547 Goto [p38](#)

4 x BC557 Goto [p38](#)

2 x BF256B Goto [p38](#)

Electrolytic Capacitors

4 x	u47/50V	Goto p39
2 x	1u/50V	Goto p39
5 x	10u/16V	Goto p39
1 x	22u/16V	Goto p40
1 x	100u/16V	Goto p40
2 x	100u/16V	Goto p40
1 x	220u/16V	

Voltage Regulators, RF choke and Inductor

1 x	7808	TO-220 8V Regulator	Goto p41
1 x	78L05	TO-92 5V Regulator	Goto p41
1 x	47uH	RF Choke	Goto p41
1 x	220uH	Inductor	

Trigger Buttons (Included in the kit)

You may wish to solder just two diagonally opposite pins of the four. This will be sufficient for the buttons to operate normally whilst allowing much easier removal and replacement should the need arise.

2 x Momentary Push Buttons **(Be sure to align the flat side with the silk screen)**

Slide Potentiometers

I have found that many of the black caps for the sliders seem to be rather loose and can easily fall off the slider levers. This is due to inconsistency in the width of the slot in the underside of the plastic cap. To solve this, apply a thin coat of clear nail lacquer to one side of the slider levers. Allow the lacquer to dry fully and test the caps on the sliders. If the caps are still loose, apply another thin coat of lacquer to the other side of the lever.

2 x	10K Audio Taper
2 x	500K Audio Taper
6 x	100K Audio Taper
6 x	100K Linear Taper

Switches

4 x	OFF-ON	Goto p42
4 x	ON-OFF-ON	Goto p42
1 x	4 position DIP switch	(Mounts on the bottom side of the board)

Connectors

1 x	3.5mm Stereo Jack Socket	(Mounts on the bottom side of the board)
2 x	14 pin (2x7) socket headers	(Mounts on the bottom side of the board)
1 x	4 pin (2x2) socket header	(Mounts on the bottom side of the board)
1 x	3 pin socket header	(Mounts on the bottom side of the board)
1 x	DC Barrel Jack socket	(Mounts on the bottom side of the board)

LED's

Before we fit the LED's we need to fit a few spacers and the top panel. This will help to ensure that the LED's sit at the correct height and position in relation to the panel.

Take 4 x 12mm M/F spacers and 4 x 20mm F/F spacers and fit them (with the 12mm spacer on the top of the board and the 20mm spacer on the bottom) in the four corner locations.

Insert the LED's into their locations (making sure that the shorter lead goes into the square pad). For the Power LED, the short leg goes closer to the 2k2 resistor. Don't solder anything yet but just let the LED's hang in position.

Place a small piece of masking tape over each LED hole on the front panel and fit the top half of the case with 4 x M3 x 5mm screws.

Turn the assembly over and adjust each LED until it sits in its panel hole and with a finger over the tape and hole, apply a little pressure so that the masking tape holds the LED in place. Cinch the leads slightly and solder in place.

Remove the masking tape and remove the panel.

IC's

1 x	TL497	
7 x	4558	Goto p43
3 x	TL082	Goto p43
2 x	MCP4822	Goto p43
1 x	6N137	Goto p43

IC's Cont.

1 x ATMEGA328

That completes the build for the Syncussion SY-1M Control Board

Board Assembly

7 x M3 x20mm F/F spacers
7 x M3 x12 mm M/F spacers
2 x M3 x12mm F/F spacers
7 x M3 x5mm M/F spacers
2 x M3 x5mm Screws

Fit 7x M3x20mm F/F spacers to the top of the Voice Board using 7x M3x5mm M/F spacers on the underside.

On the top side of the Control Board Fit 2 x M3 x12mm F/F spacers to the 2 middle holes on the long sides using 2 x M3 x5mm screws.

Take the breakaway pin strips provided with the kit and break off;

4 x 7 pins, 2 x 2 pins and 1 x 3pins

Please note that the Pin Connectors provided with the kit are slightly too long. With a pair of sharp snips, trim approximately 2mm from each end of the pins.

Fit the pin strip to either the Control Board headers or the Voice board headers

The two 2 pin pieces are for the square 4 way header.

Plug the boards together.

Fit 7 x M3 x12mm M/F spacers to the remaining locations on the top of the Control Board thus securing the boards together.

Calibration

NOISE

Power up the unit and measure the noise level at point 3

The original service manual recommends adjusting the noise level to 2.5V p-p however, transistors may vary greatly in the level of noise produced so you may not achieve the full 2.5Vp-p. If you cannot adjust the level at point 3 to be much more than 2Vp-p you may increase the level by removing the 10k resistor just below the trimmer.

VC02 OFFSET (This is described as it is in the original service manual but don't worry about being too precise with this adjustment, It can also be a matter of preference and different offsets will give different results.)

With the power ON set Ch.1 controls as follows:

TUNE - 5
SWEEP - OFF
L.F.O - OFF
S/H - OFF

Connect a frequency counter or tuner to Channel 1, point 1 on the underside of the voice board and adjust Ch1 TUNE so that the counter reads 440Hz.

Connect your frequency counter to point 2 on the underside of the voice board and adjust the VC02 OFFSET trimmer so that the counter reads 450Hz.

Repeat the VC02 OFFSET calibration for Channel 2.

The next two calibrations for the scale tuning should be done with OSC. MODE set to "A" on both channels, with "SWEEP", "L.F.O" and "S/H" switches all in the "OFF" position.

CV -V/OCT

Plug a 1V/Oct source into Ch.1 "Tune" input and adjust the Ch.1 "CV-V/Oct" trimmer to give a doubling of frequency (1 Octave) at the output for a 1 Volt increase at the "Tune" input.

Repeat for channel 2

MIDI-V/OCT (Details of the required MIDI connector are given in the MIDI Document.)

First, a word on the way the SY-1M receives incoming MIDI data.

The DIP switches for setting the MIDI channel always set the MIDI channel for Ch.1

Ch.2 always receives on the next highest channel to that set for Ch.1

For example; If Ch.1 is set to receive on MIDI channel 5, then Ch. 2 will receive on MIDI channel 6 or, if Ch.1 is set to receive on MIDI channel 11, then Ch. 2 will receive on MIDI channel 12.

However, if Ch.1 is set to receive on MIDI channel 16, then Ch.2 will receive on MIDI channel 15.

Refer to the MIDI Document for DIP switch reference chart.

NOTE: The MIDI receive channel is always set on power up so to change channels you need to power down, set the receive channel and power up again.

So, with the unit powered OFF, connect a MIDI source (DAW/ Synth/Controller Keyboard) to the MIDI input.

Switch the MIDI MODE to “2” and set the MIDI channel to “1”, (All DIP switches in the “0” position)

Set your MIDI source to transmit on MIDI Channel 1

Power up the SY-1M

Send alternating notes 2 octaves apart and adjust Ch.1 “MIDI- V/OCT” until you have exactly 2 Octaves between notes. (you may do this over 3 octaves for greater precision)

Set your MIDI source to transmit on MIDI Channel 2 and repeat the process for Ch.2 (The DIP switches should of course still be set to receive on MIDI Channel 1)

That’s it! Let’s get it boxed up.

Final Assembly

Remove all nuts and washers from the I/O connectors.

Slide the board assembly into the bottom half of the case and secure to the base with the 7x M3x3mm screws provided with the kit.

Fit the I/O connector nuts.

Slowly lower the top half of the case onto the assembly making sure that everything is sitting squarely in its panel cut-out.

Fit 9x M3x5mm screws to the top of the front panel .

Fit the 4 remaining M3x5mm screws to the case sides.

Fit the 4 adhesive rubber feet

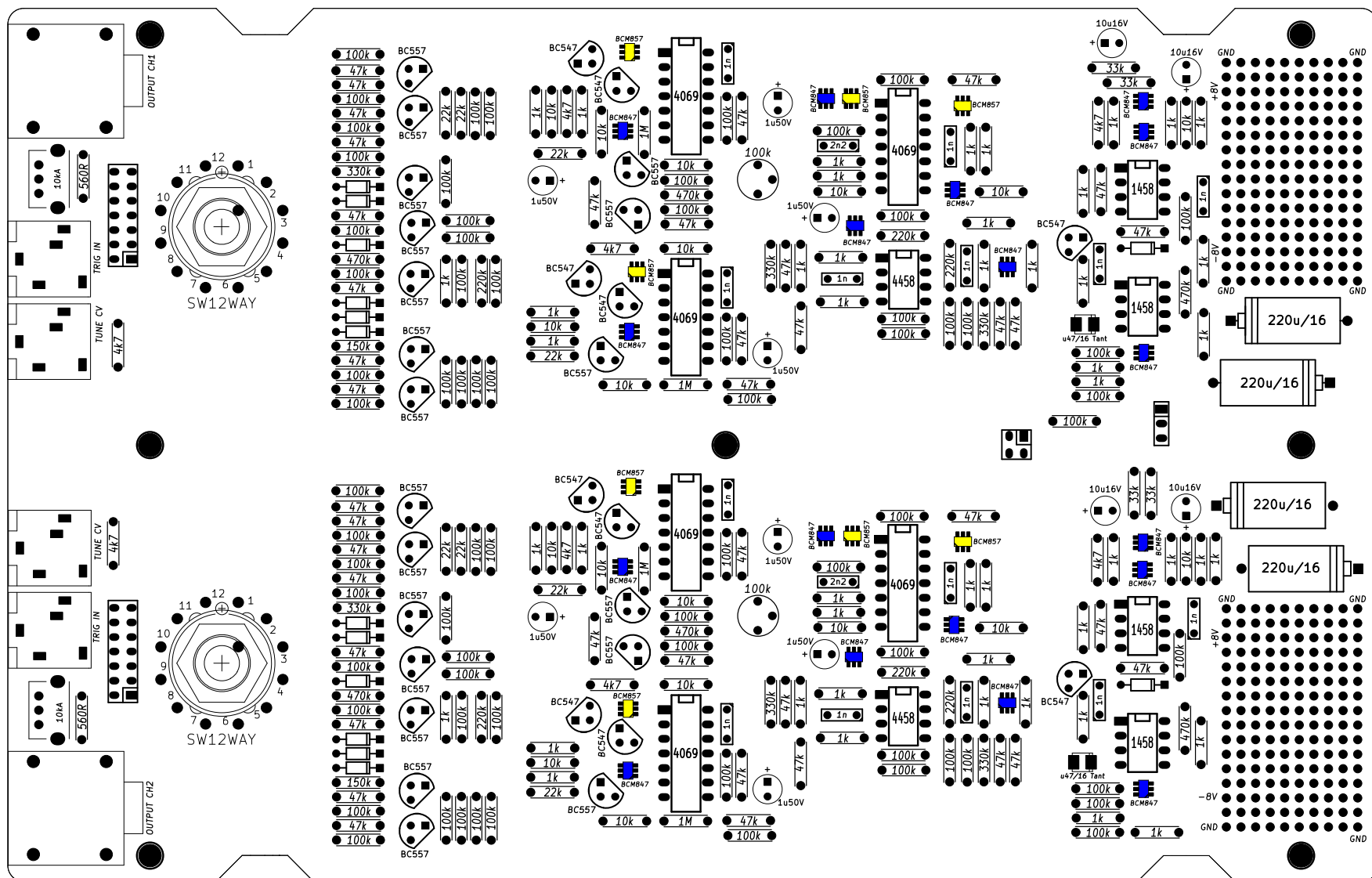
Fit the 16 Black slider caps and the Rotary Switch Knob.

8 x

BCM857

18 x

BCM847



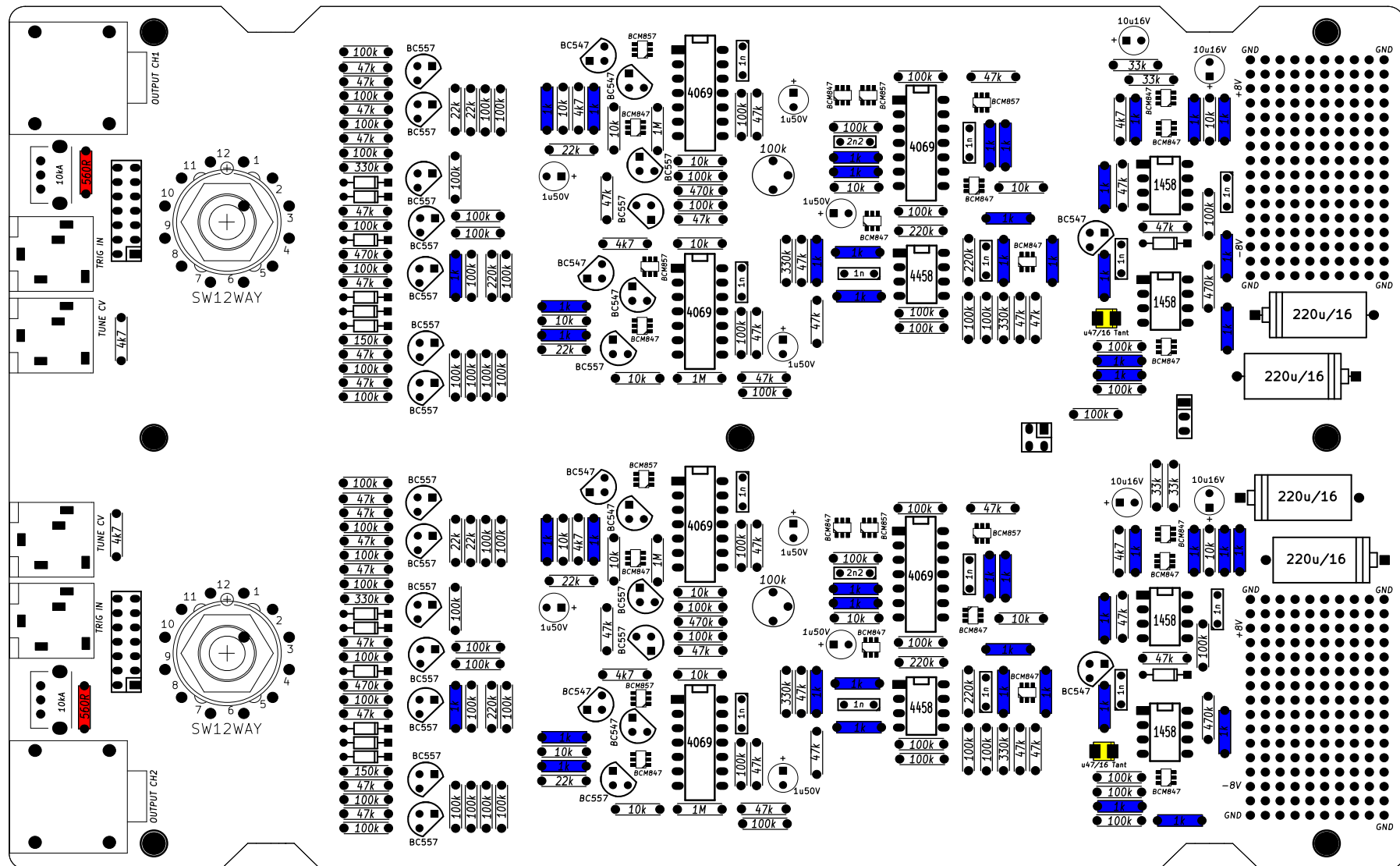
2x

u47/35V

Resistors

2 x 560R

48 x 1k

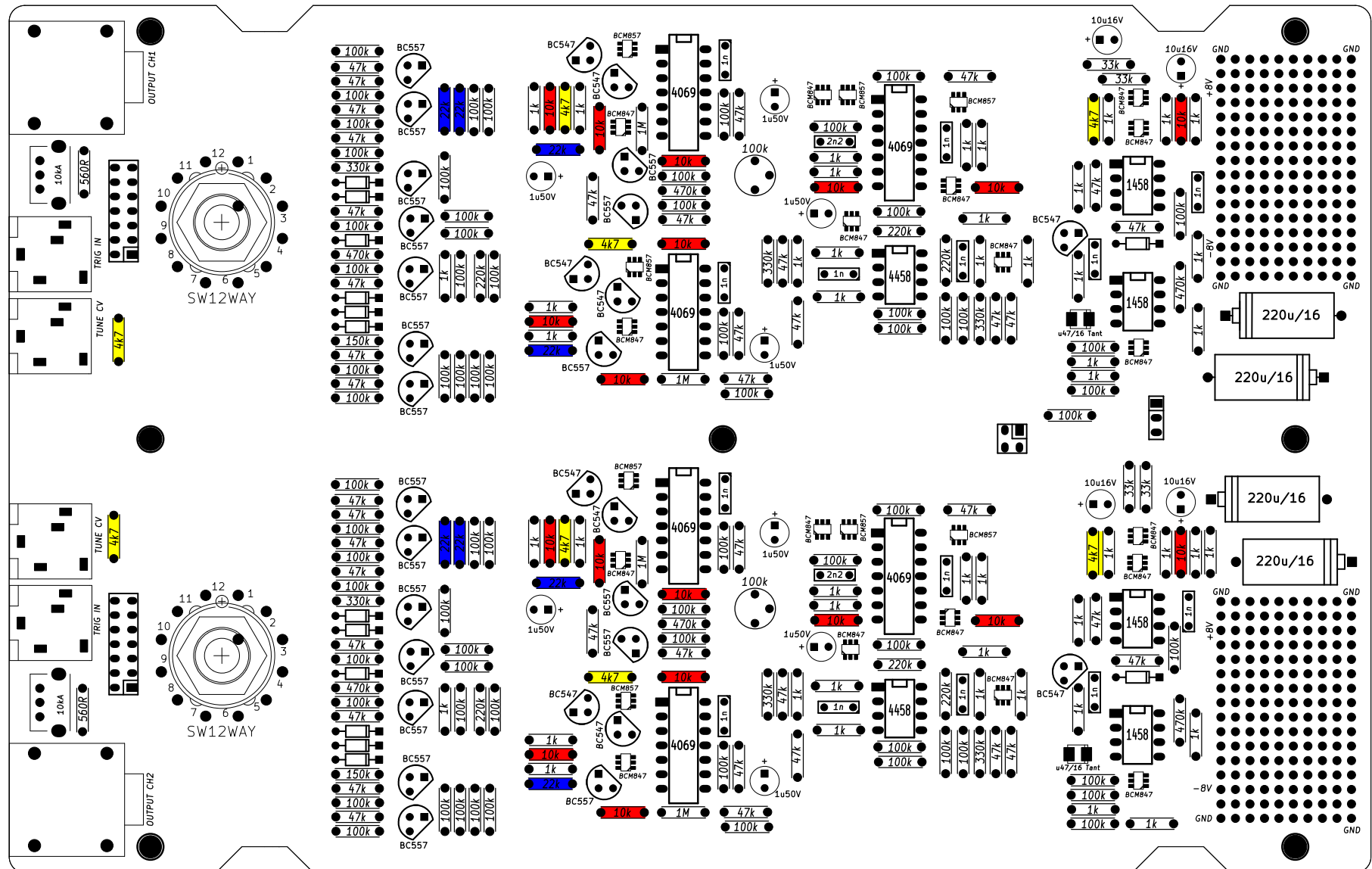


Resistors

8x 4k7

18x 10k

8x 22k

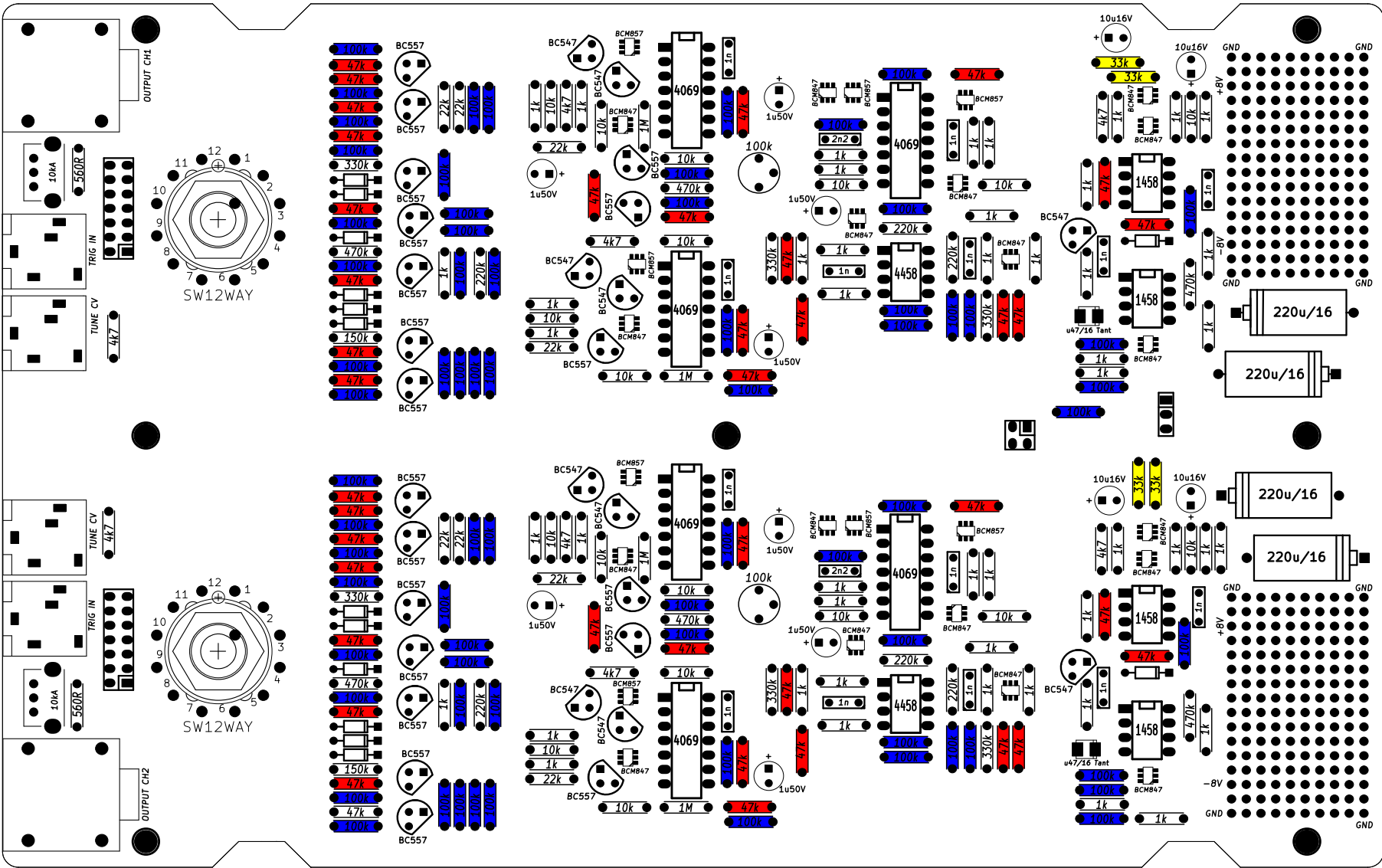


Resistors

4x 33k

40x 47k

70x 100k

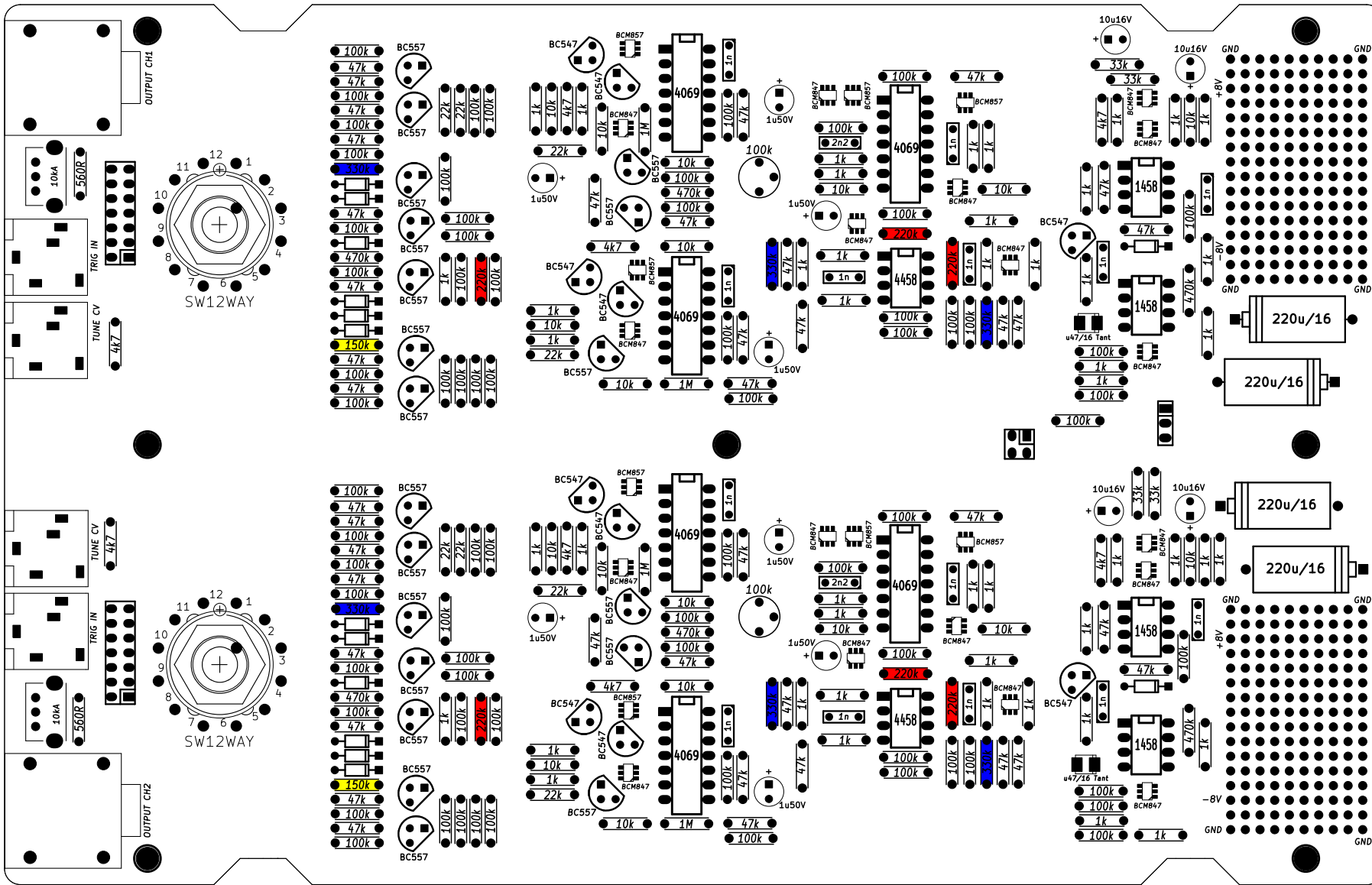


Resistors

2x 150k

6x 220k

4x 330k

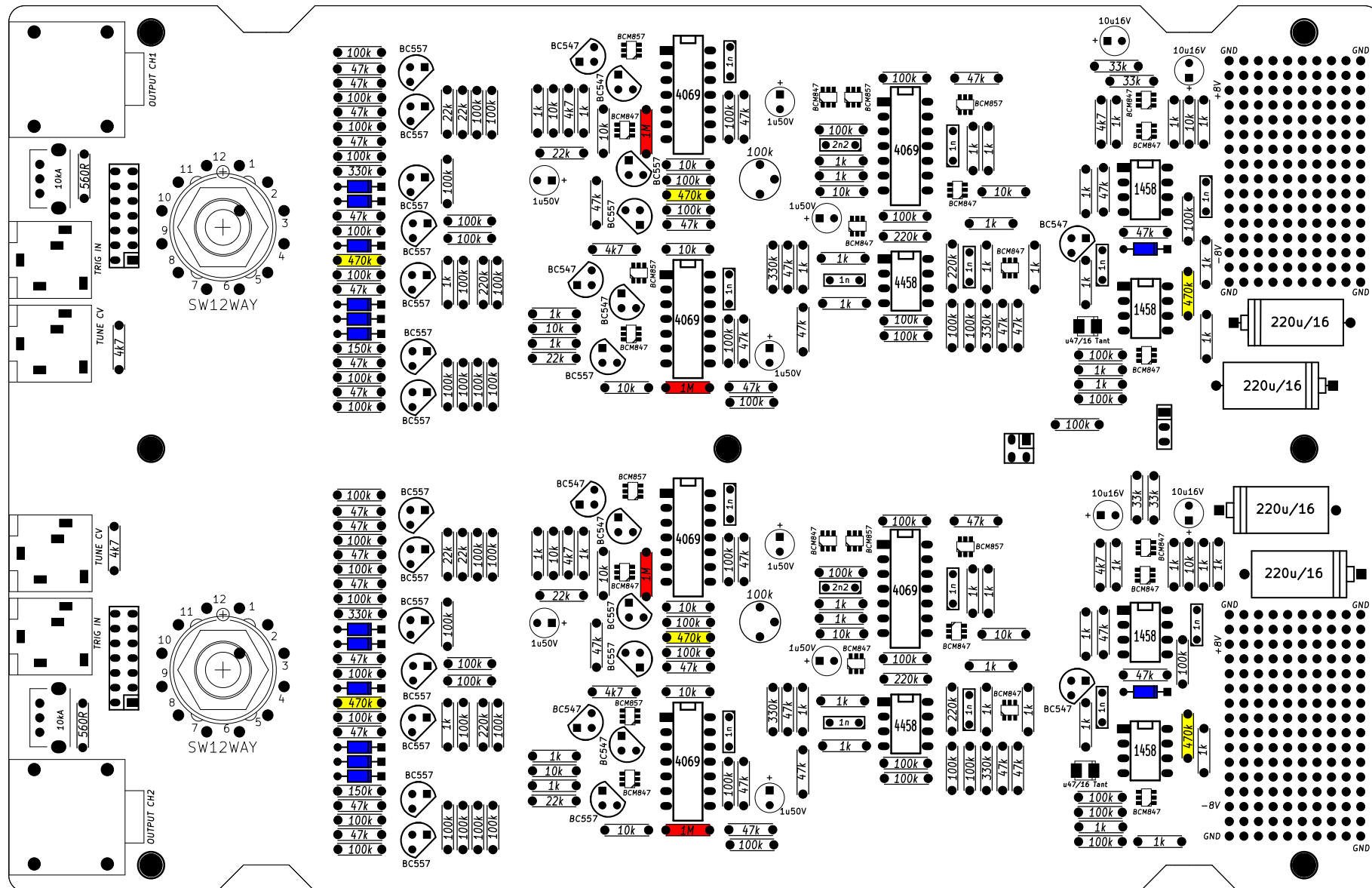


Resistors

6x 470k

4x 1M

14x Diodes

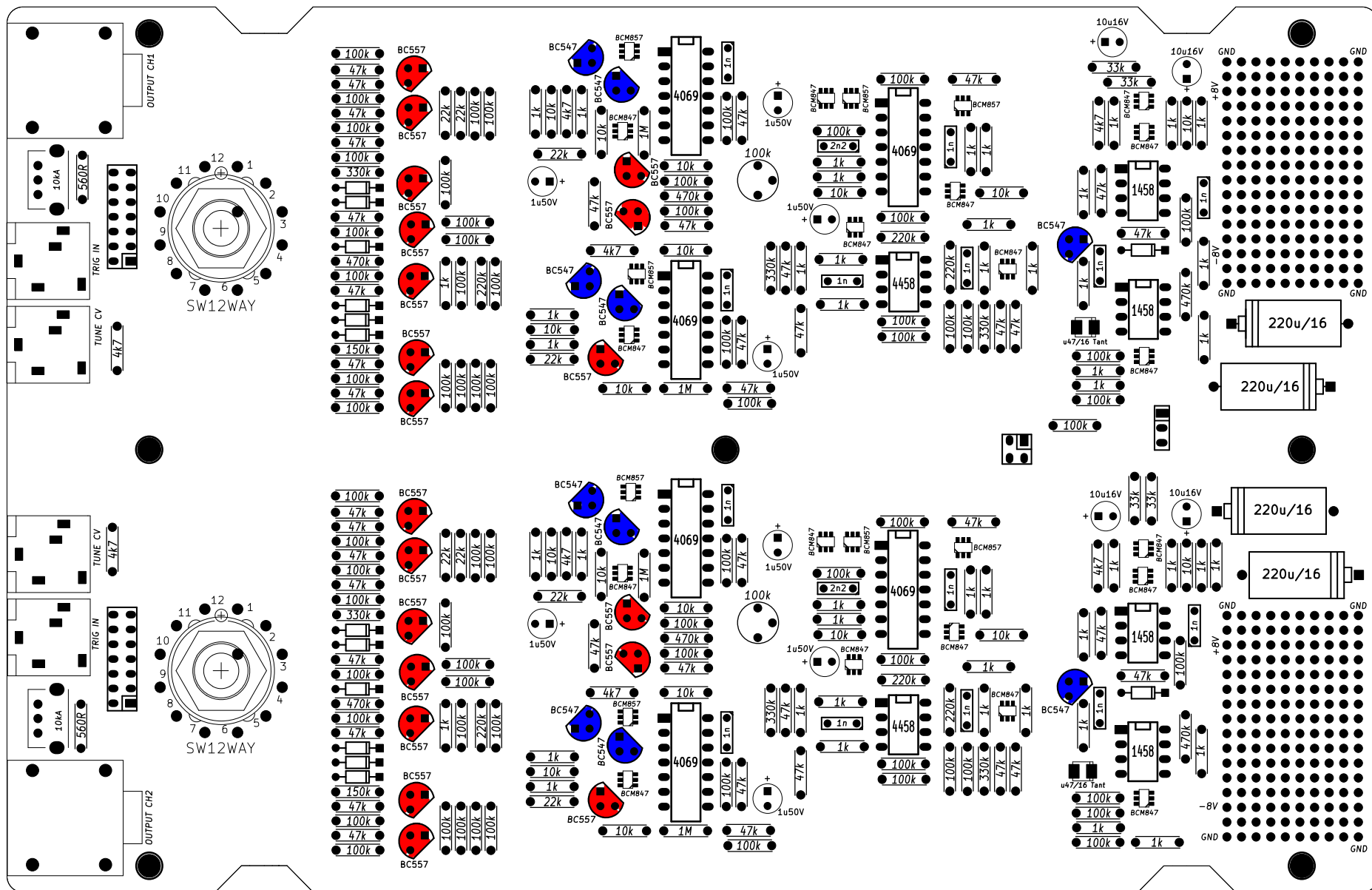


$2 \times 2n^2$ 

T0-92 Transistors

10 x BC547

20 x BC557

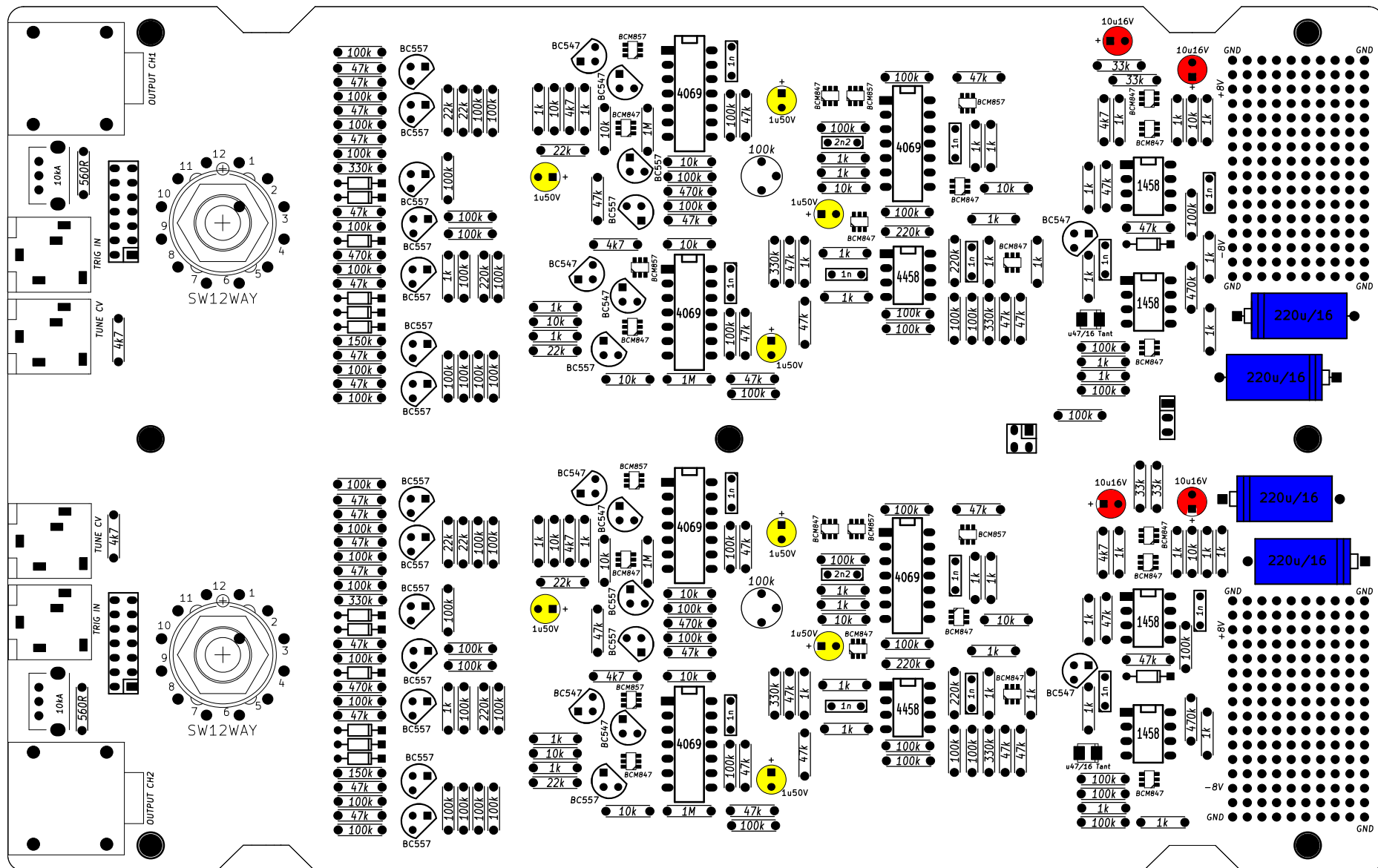


Al Electrolytic Capacitors

8 x 1u/50V

4 x 10u/16V

4 x 220u/16V

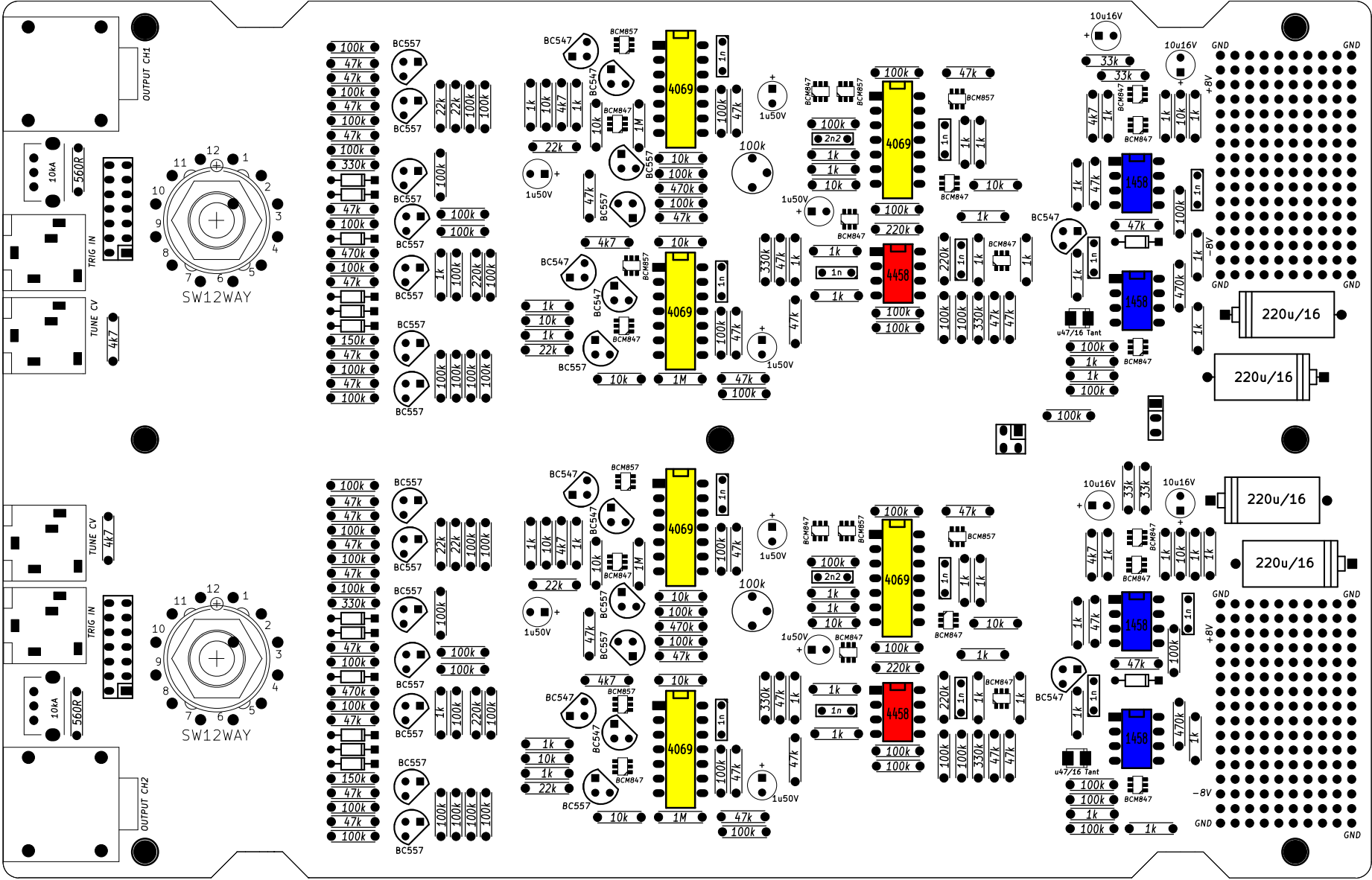


IC's

6 x CD4069

2 x 4458

4 x 1458



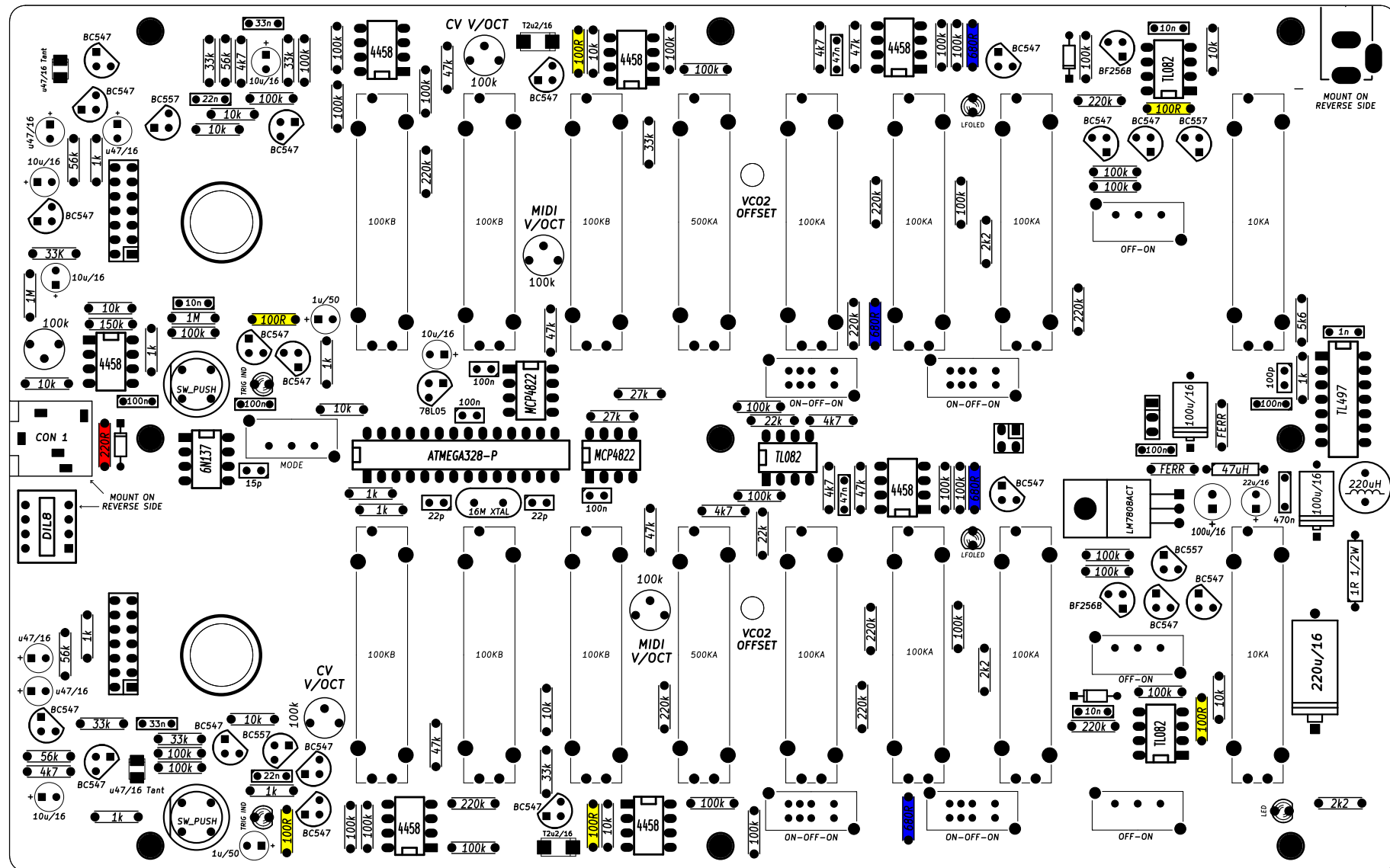
$2 \times 2u^2/35V$ 

Resistors

6 x100R

1 x220R

4 x680R

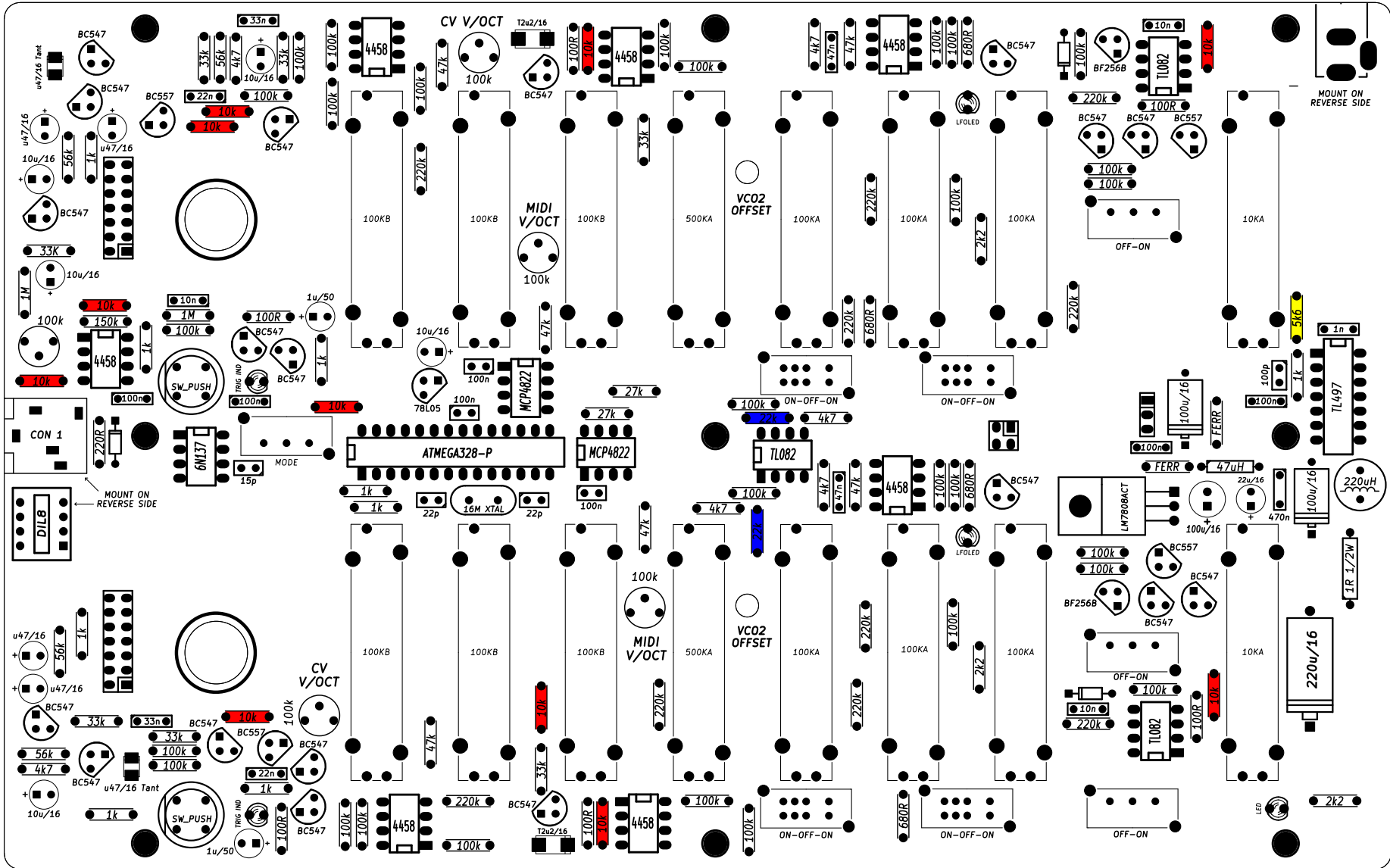


Resistors

1 x 5k6

11 x 10k

2 x 22k

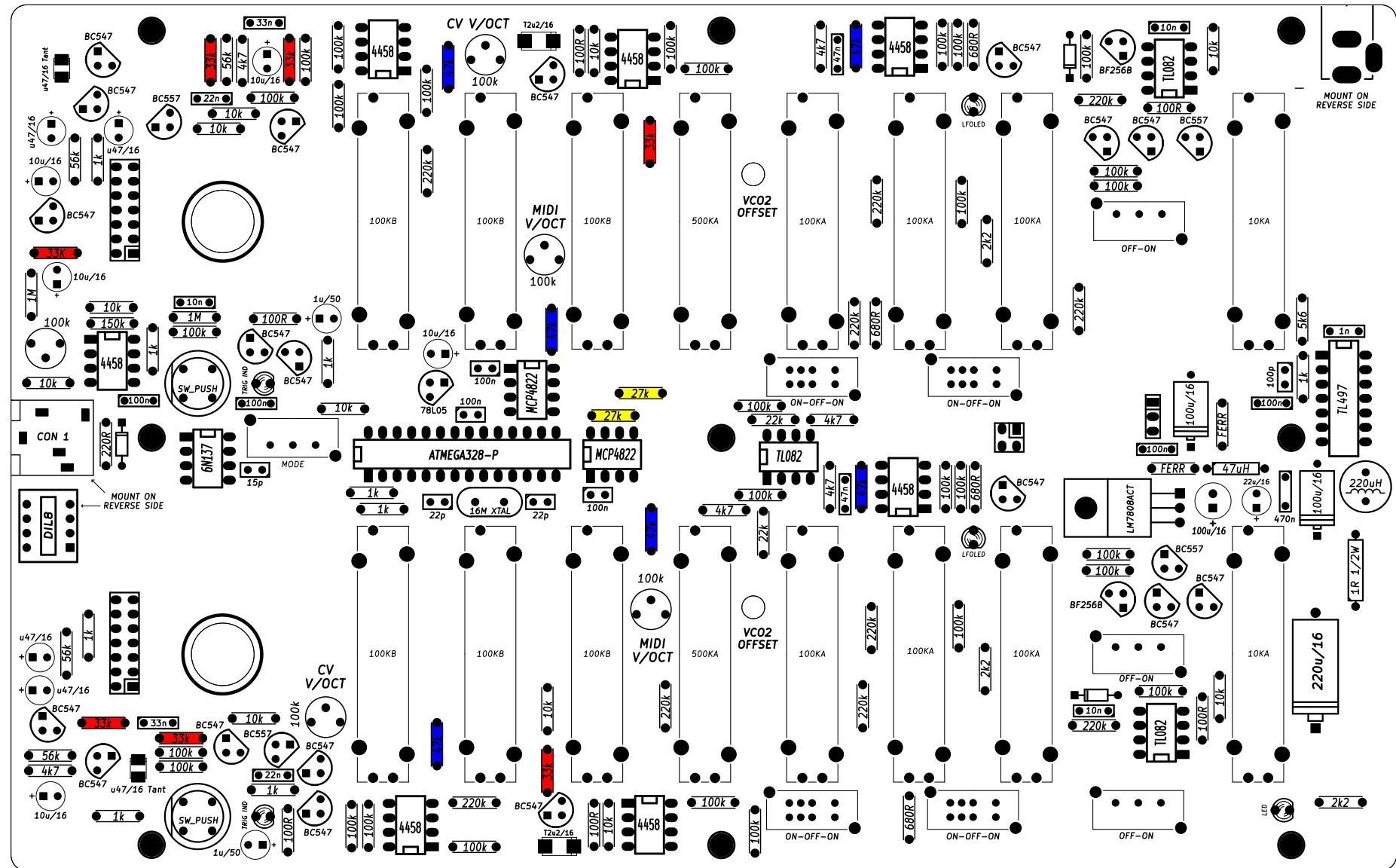


Resistors

2 x 27K

7 x 33K

6 x47K

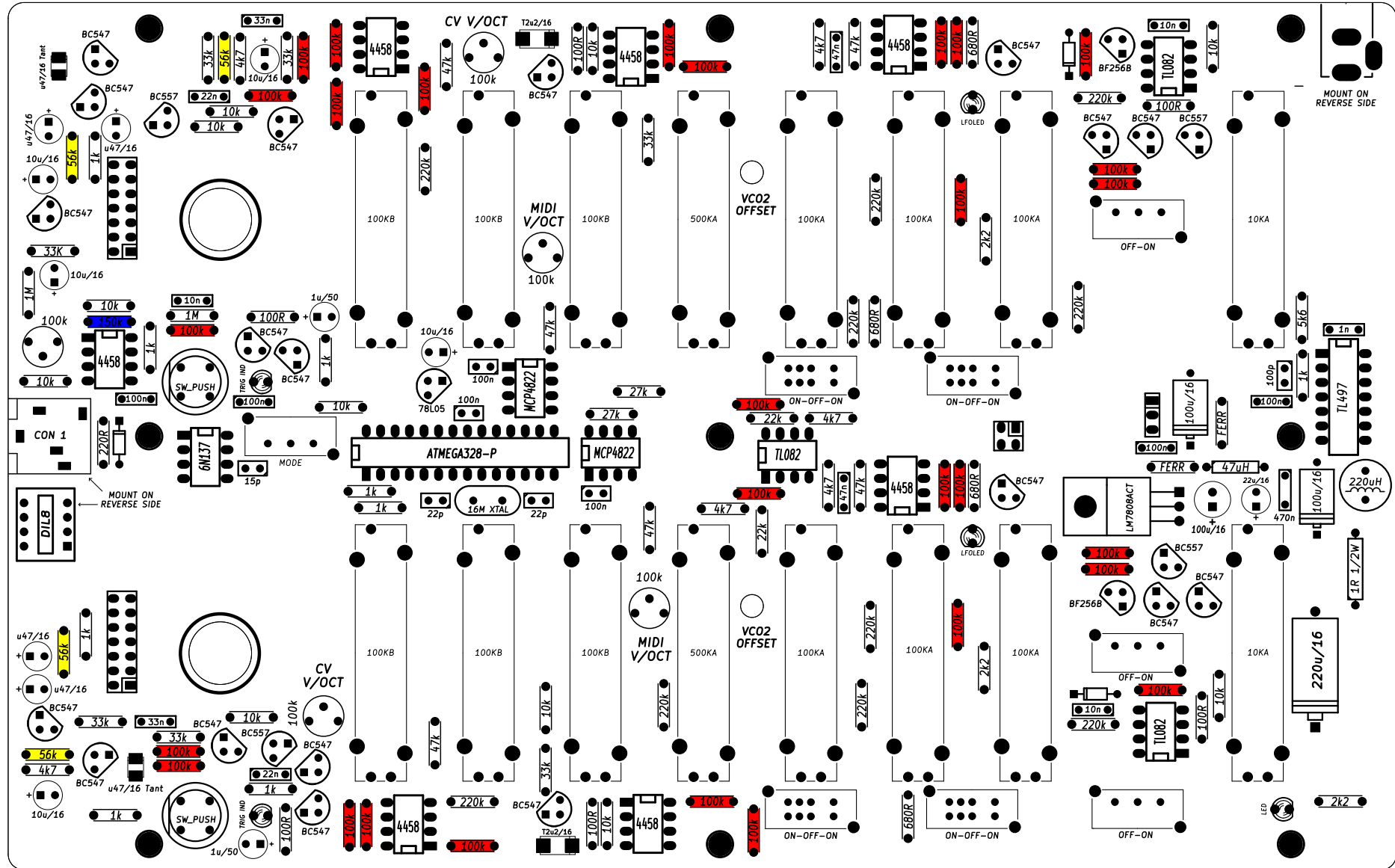


Resistors

4 x56K

29 x 100K

1x150K

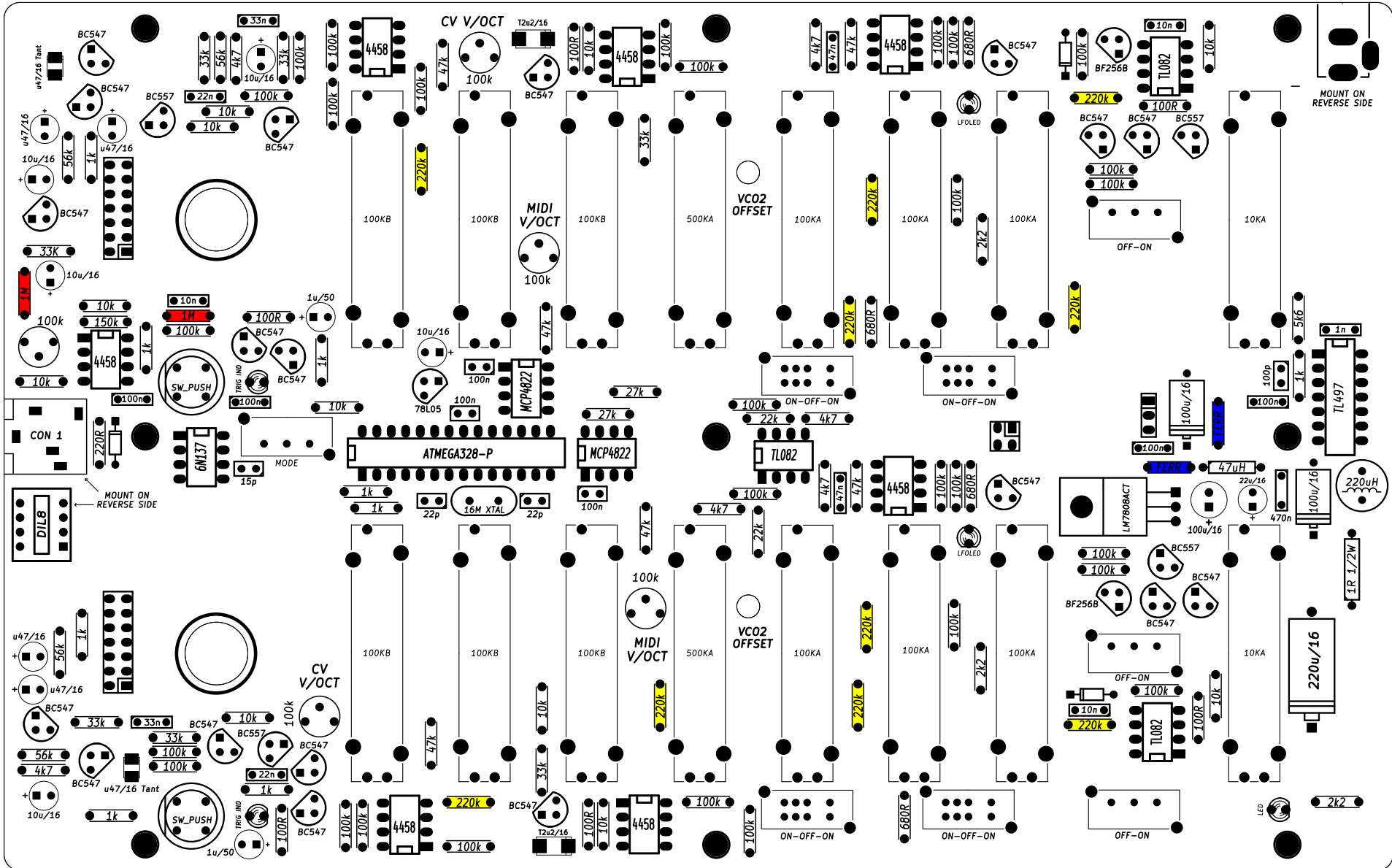


Resistors

10 x 220K

2 x 1M

2 x Ferrite Beads

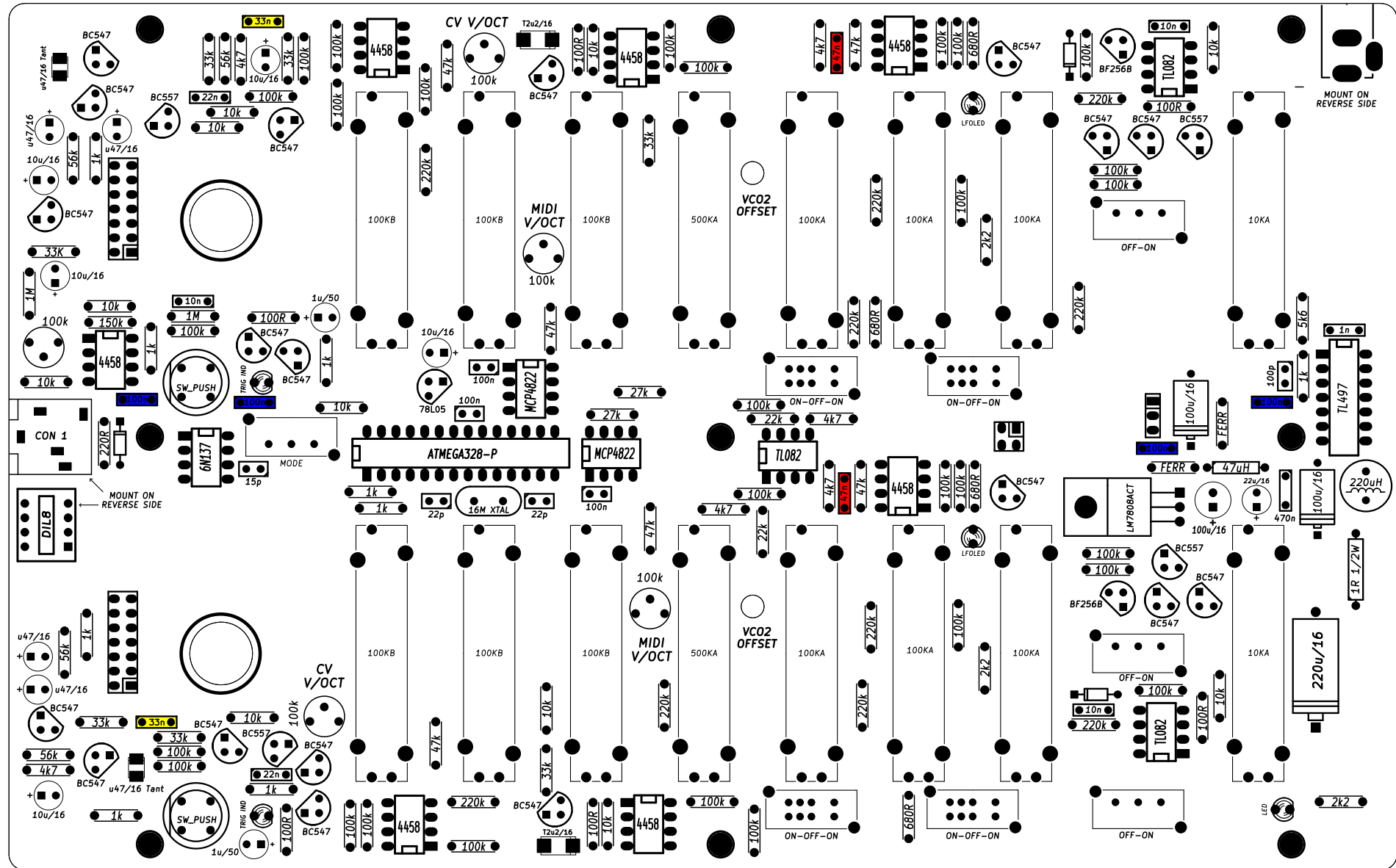


Film Capacitors

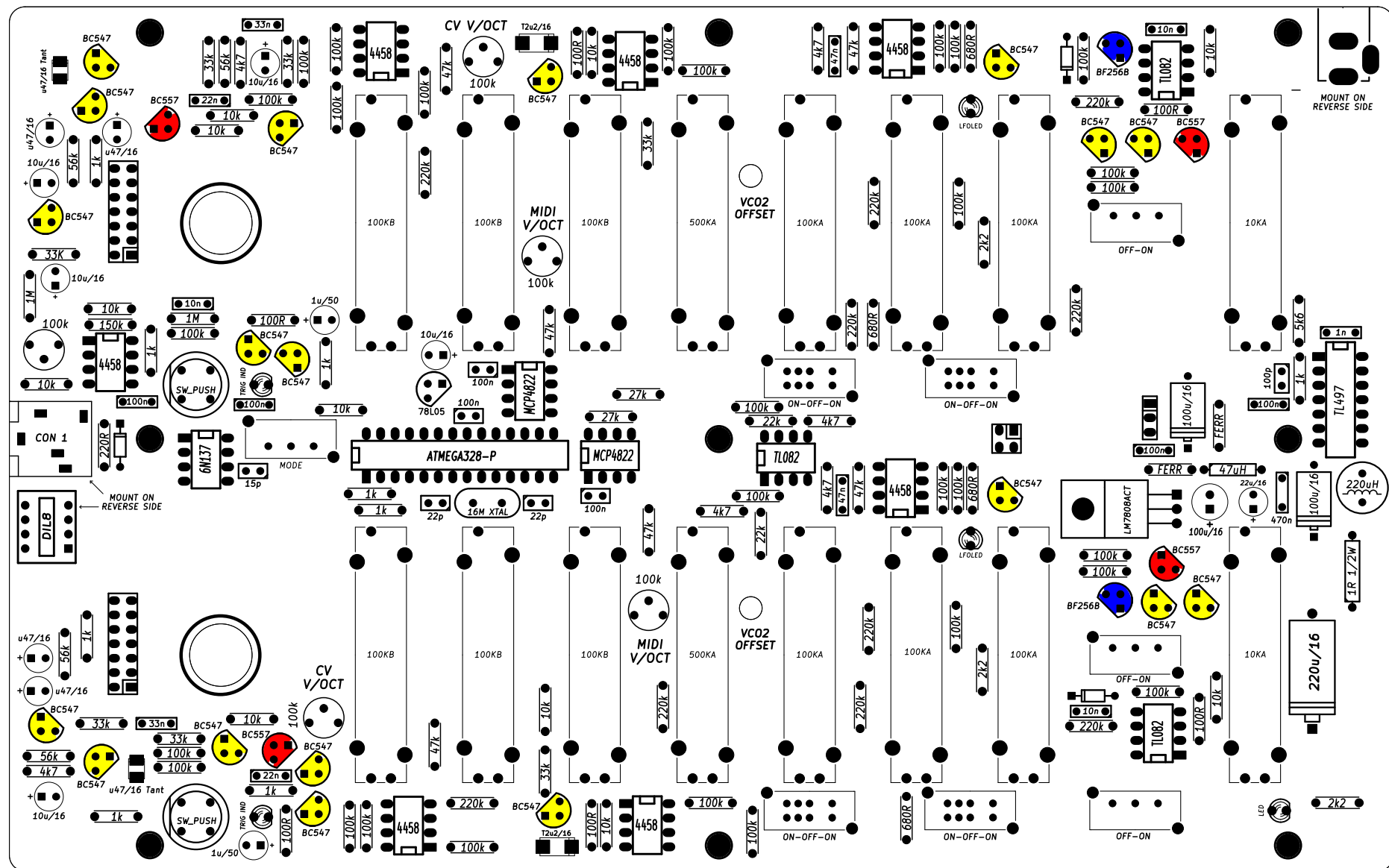
2 x 33n

2 x 47n

4 x 100n



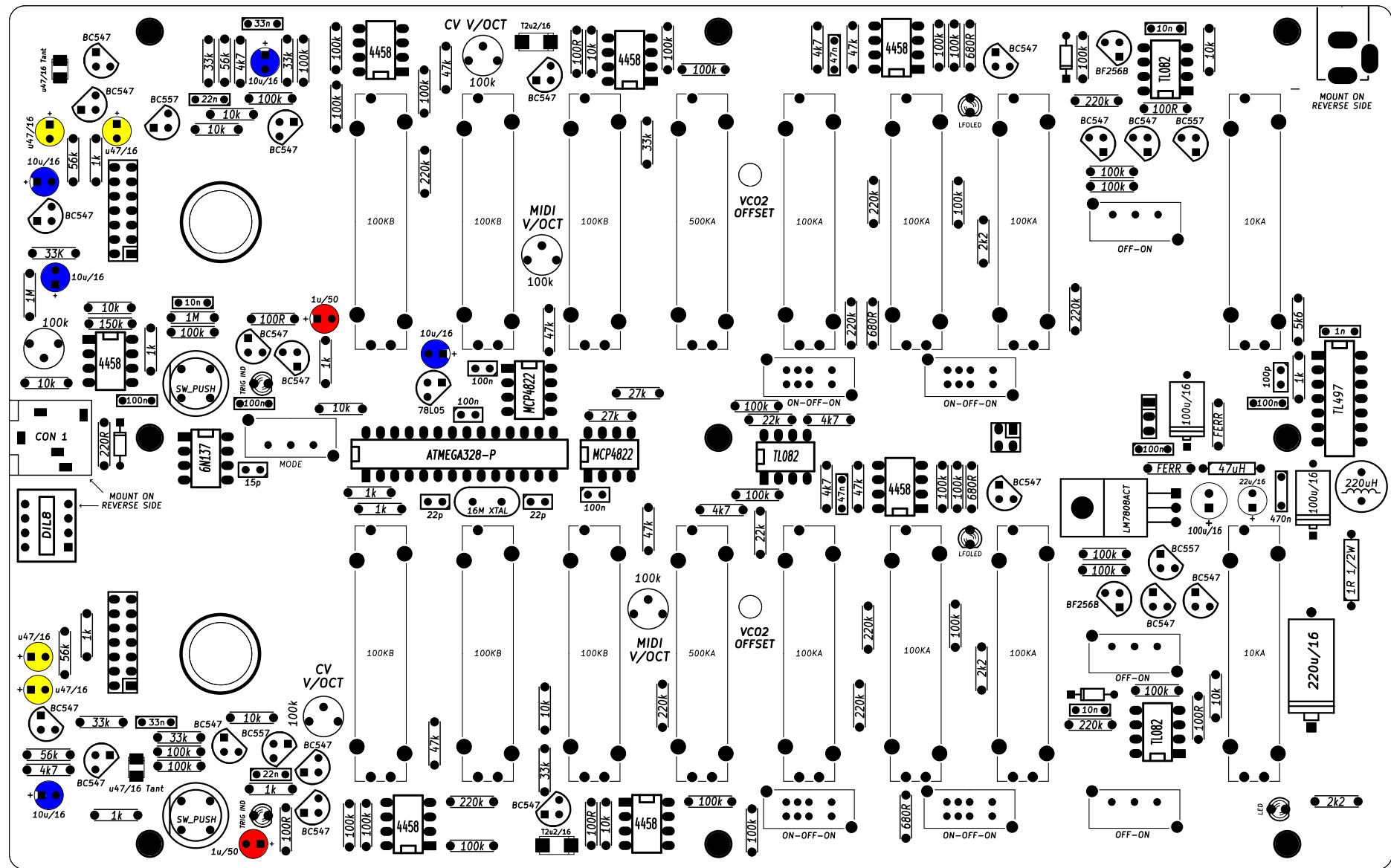
Transistors 19 x BC547 4 x BC557 2 x BF256B



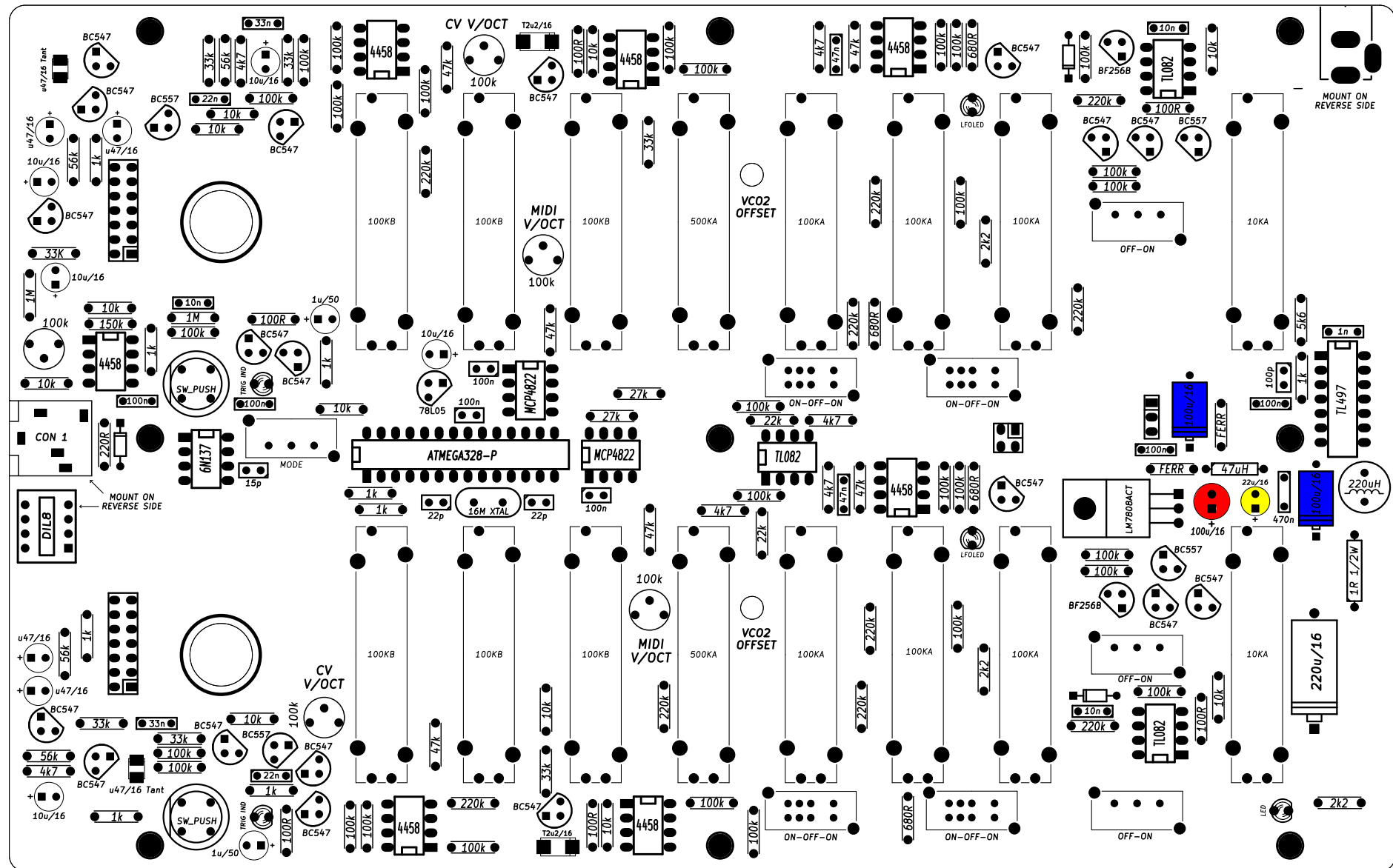
Al Electrolytic Capacitors

2 x 1u/50V

5 x 10u/16V



2 x 100u/16V

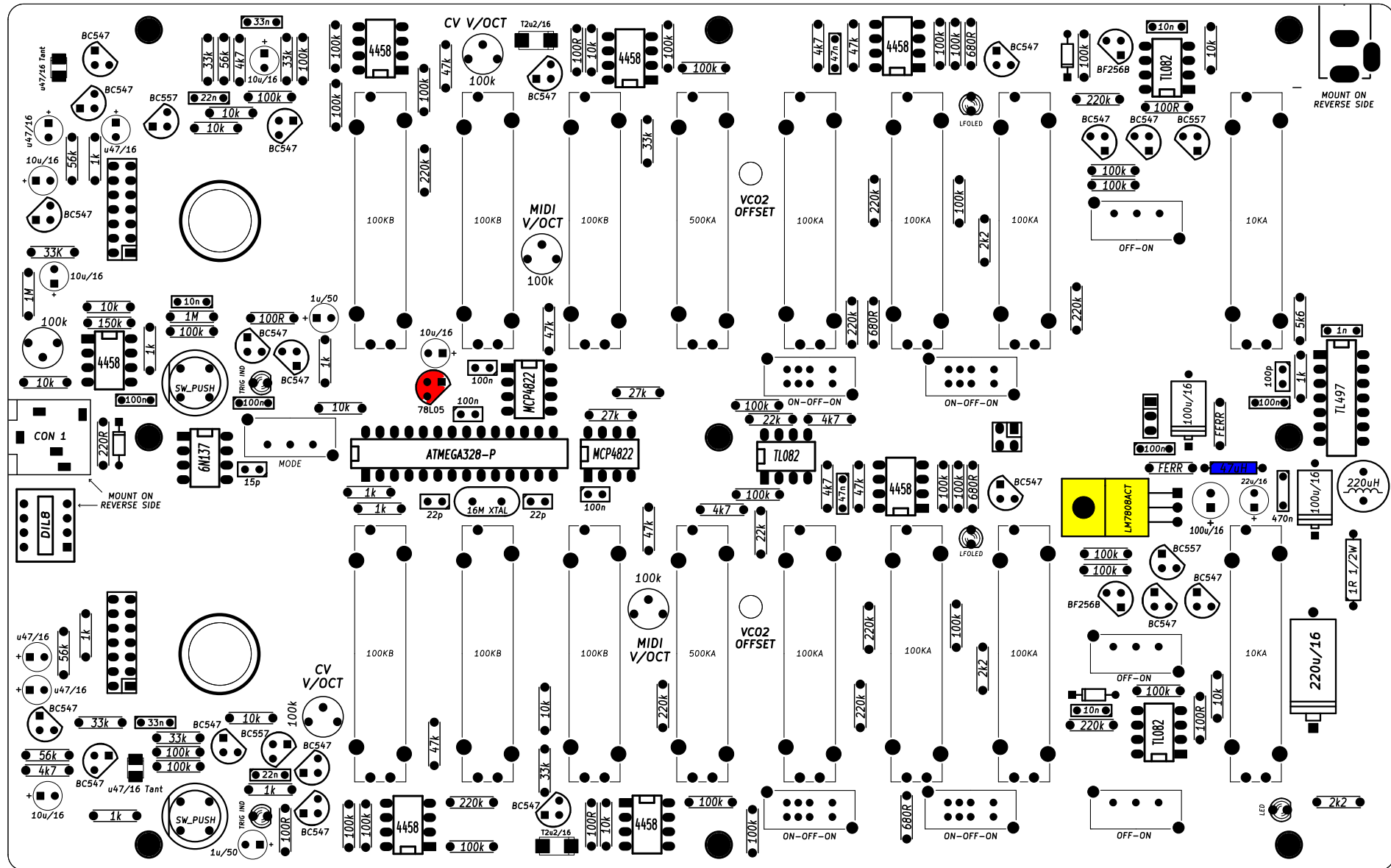


Voltage Regulators RF choke

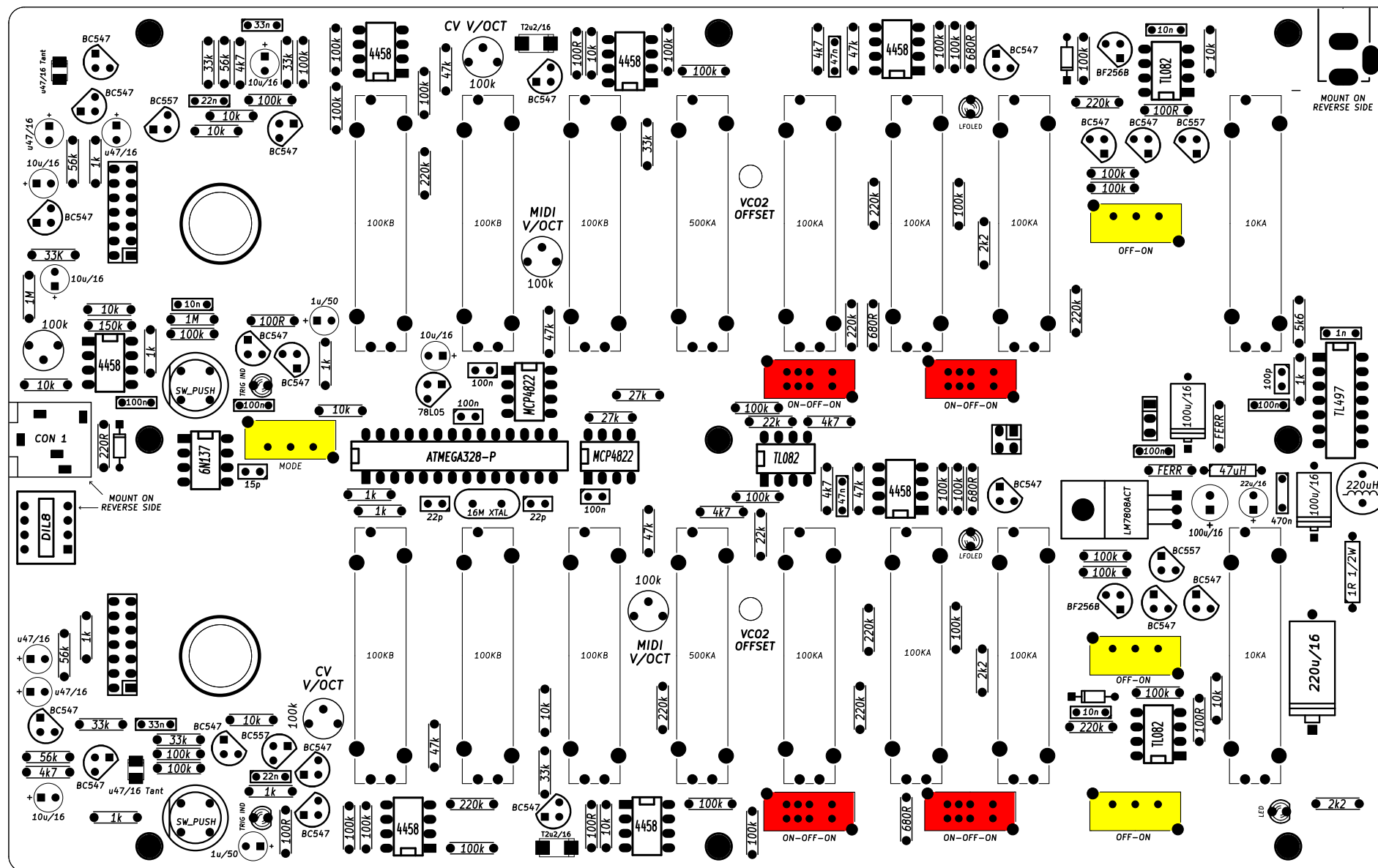
1x7808

1x78L05

1x47uH



4 x ON-OFF-ON



IC's

7 x 4558

3 xTL082

2 x MCP4822

1 x6N137

